BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

Name: Evan Wu

ID: 5839048

> **Due Date**
>
> Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\evan\Documents\GitHub\hw3-evtw`
```

```julia
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day-1, and the decay rates of CBOD and NBOD are are 0.35 and 0.25 day-1, respectively. The river's saturated dissolved oxygen concentration is 10m g/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance `d` km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m3/d	10,000 m3/d	15,000 m3/d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

Streeter-Phelps solution:

$$U\frac{dC}{dx} = k_a(C_s - C) + P - R - S_B - k_c B_0 exp(\frac{-k_c x}{U}) - k_n N_0 exp(\frac{-k_n x}{U})$$

In this model, we ignore P, R, S_B.

$$U\frac{dC}{dx} = k_a(C_s - C) - k_c B_0 exp(\frac{-k_c x}{U}) - k_n N_0 exp(\frac{-k_n x}{U})$$

From the known variables:

$$k_a = \frac{0.55}{day}$$

$$k_c = \frac{0.35}{day}$$

$$k_n = \frac{0.25}{day}$$

$$U = \frac{6km}{day}$$

$$C_s = 10\frac{mg}{L}$$

Need to split the system into two parts: the stream before Stream 2 is integrated and the river after Stream 2 is added:

Integrating the equation:

$$C(x) = C_s(1-\alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$

$$\alpha_1 = exp(-\frac{k_a x}{U})$$

$$\alpha_2 = (\frac{k_c}{k_a - k_c})[exp(-\frac{k_c x}{U}) - exp(-\frac{k_a x}{U})]$$

$$\alpha_3 = (\frac{k_n}{k_a - k_n})[exp(-\frac{k_n x}{U}) - exp(-\frac{k_a x}{U})]$$

Need to find the initial concentration C_0 when $x = 0$. Let $x = 0$ when the river and Stream 1 just mix:

$$\alpha_1 = e^{-0} = 1$$

$$\alpha_2 = (\frac{k_c}{k_a - k_c})(1-1) = 0$$

$$\alpha_3 = (\frac{k_n}{k_a - k_n})(1-1) = 0$$

$$C(x) = C_0$$

Initial concentration of C_0, B_0, N_0:

$$C_0 = 7.5\frac{mg}{L} \times \frac{10}{11} + 5\frac{mg}{L} \times \frac{1}{11} \approx 7.27\frac{mg}{L}$$

$$B_0 = 5\frac{mg}{L} \times \frac{10}{11} + 50\frac{mg}{L} \times \frac{1}{11} \approx 9.09\frac{mg}{L}$$

$$N_0 = 5\frac{mg}{L} \times \frac{10}{11} + 35\frac{mg}{L} \times \frac{1}{11} \approx 7.73\frac{mg}{L}$$

Next, we need to consider the system after Stream 2 is added when x = 15km. First, the concentration of the river before Stream 2 is added must be calculated:

$$C(x=15) = C_s(1-\alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$

$$\alpha_1 = e^{-\frac{0.55\times15}{6}} \approx 0.252$$

$$\alpha_2 = \frac{0.35}{0.55-0.35}(e^{-\frac{0.35\times15}{6}} - e^{-\frac{0.55\times15}{6}}) \approx 0.287$$

$$\alpha_3 = \frac{0.25}{0.55-0.25}(e^{-\frac{0.25\times15}{6}} - e^{-\frac{0.55\times15}{6}}) \approx 0.235$$

Calculating the concentration of the river before Stream 2 is added:

$$C(x=15) = 10(1-0.252) + 7.27(0.252) - 9.09(0.287) - 7.73(0.235) \approx 4.89\frac{mg}{L}$$

New system equation after Stream 2 is added:

$$C(x) = C_s(1-\alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$

$$\alpha_1 = exp(-\frac{k_a(x-15)}{U})$$

$$\alpha_2 = (\frac{k_c}{k_a-k_c})[exp(-\frac{k_c(x-15)}{U}) - exp(-\frac{k_a(x-15)}{U})]$$

$$\alpha_3 = (\frac{k_n}{k_a-k_n})[exp(-\frac{k_n(x-15)}{U}) - exp(-\frac{k_a(x-15)}{U})]$$

We subtract x by 15 as the mixing starts at x = 15. The initial concentrations after the river and Stream 2 need to be calculated:

$$C_0 = 4.89\frac{mg}{L} \times \frac{110}{125} + 5\frac{mg}{L} \times \frac{15}{125} \approx 4.90\frac{mg}{L}$$

$$B_0 = 9.09(1-0.287)\frac{mg}{L} \times \frac{110}{125} + 45\frac{mg}{L} \times \frac{15}{125} \approx 11.10\frac{mg}{L}$$

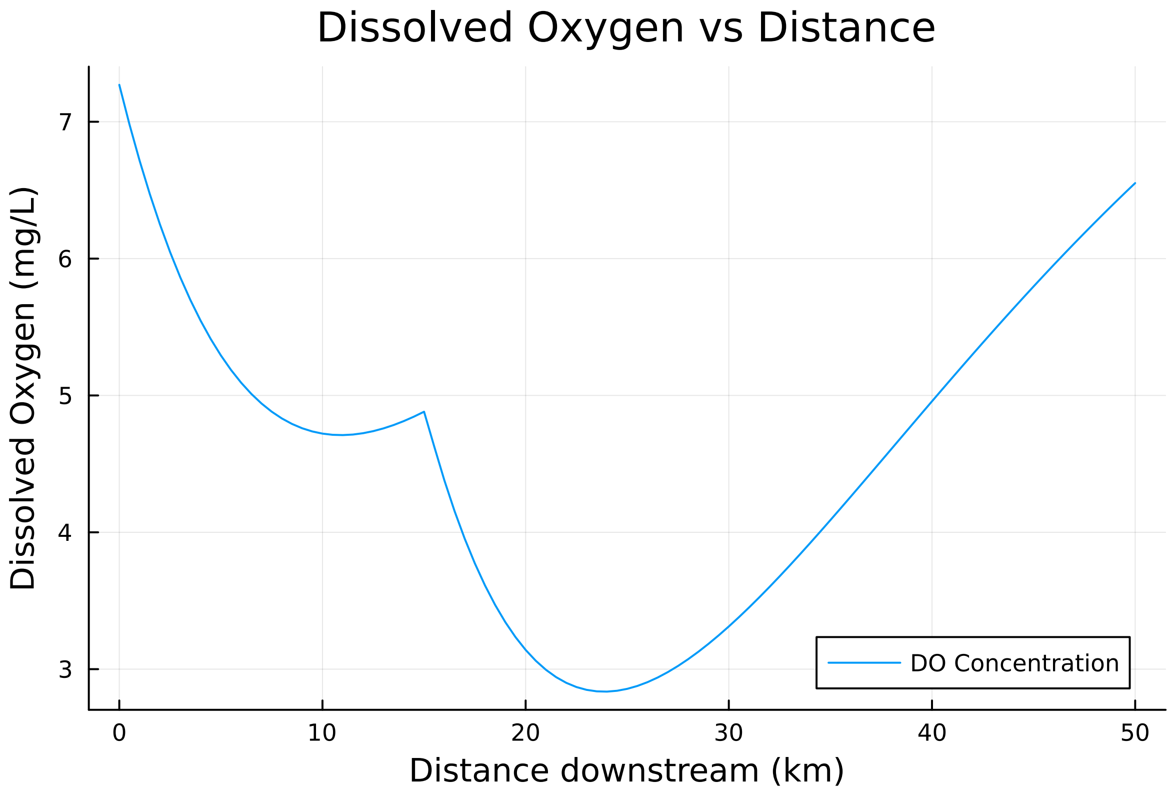
$$B_0 = 7.73(1-0.235)\frac{mg}{L} \times \frac{110}{125} + 35\frac{mg}{L} \times \frac{15}{125} \approx 9.40\frac{mg}{L}$$

With these two system equations, it is possible to plot the dissolved oxygen concentration over distance downstream

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
```

```
U = 6 # km/d

# System 1:
C1_0 = 7.27 # mg/L
B1_0 = 9.09 # mg/L
N1_0 = 7.73 # mg/L

# System 2:
C2_0 = 4.90 # mg/L
B2_0 = 11.10 # mg/L
N2_0 = 9.40 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
```

```
2.8358280713478363
```



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

Numerical integrated solution will be utilizing the Forward Euler discretization.

$$U\frac{dC}{dx} = k_a(C_s - C) - k_c B_0 exp(\frac{-k_c x}{U}) - k_n N_0 exp(\frac{-k_n x}{U})$$

$$\frac{dC}{dx} = \frac{k_a}{U}(C_s - C) - \frac{k_c}{U} B_0 exp(\frac{-k_c x}{U}) - \frac{k_n}{U} N_0 exp(\frac{-k_n x}{U})$$

$$As\frac{dC}{dx} \approx \frac{\Delta C(x)}{\Delta x} = \frac{C(x+\Delta t) - C(x)}{\Delta x}$$

$$\frac{C(x+\Delta t) - C(x)}{\Delta x} = \frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U} B_0 exp(\frac{-k_c x}{U}) - \frac{k_n}{U} N_0 exp(\frac{-k_n x}{U})$$

$$C(x+\Delta t) - C(x) = \Delta x[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 exp(\frac{-k_c x}{U}) - \frac{k_n}{U}N_0 exp(\frac{-k_n x}{U})]$$

$$C(x+\Delta t) = C(x) + \Delta x[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 exp(\frac{-k_c x}{U}) - \frac{k_n}{U}N_0 exp(\frac{-k_n x}{U})]$$

For t = 0.5,

$$C(x+0.5) = C(x) + 0.5[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 exp(\frac{-k_c x}{U}) - \frac{k_n}{U}N_0 exp(\frac{-k_n x}{U})]$$

When Stream 2 is added (x >= 15), the equation will change:

$$C_0 = C(x=15) * \frac{110}{125} + 5 * \frac{15}{125}$$

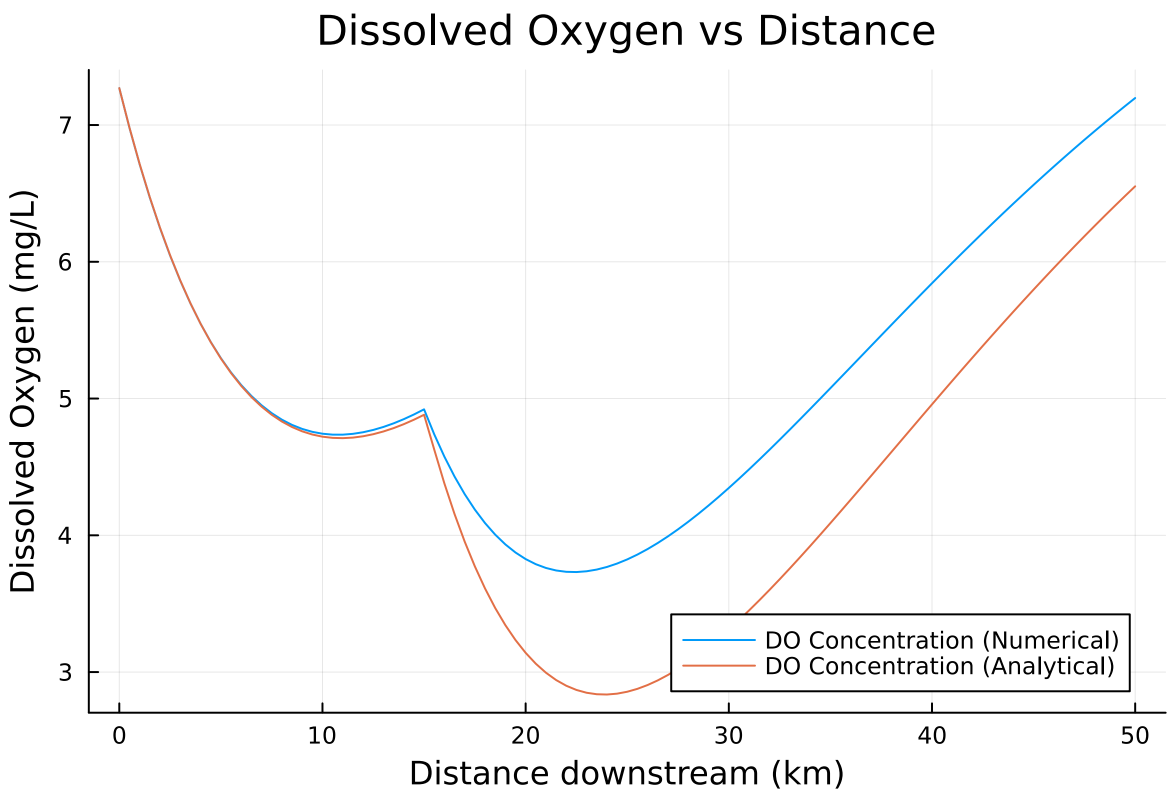
$$B_0 = B(x=15) * \frac{110}{125} + 45 * \frac{15}{125}$$

$$N_0 = N(x=15) * \frac{110}{125} + 45 * \frac{15}{125}$$

$$C(x+\Delta t) = C(x) + \Delta x[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 exp(\frac{-k_c(x-15)}{U}) - \frac{k_n}{U}N_0 exp(\frac{-k_n(x-15)}{U})]$$

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

function simulate_timestep1(C, i, C0, B0, N0)
    return k_a/U*(C_s - C) - k_c/U*B0*exp(-k_c*i/U) - k_n/U*N0*exp(-k_n*i/U)
end

function simulate_timestep2(C, i, C0, B0, N0)
    return k_a/U*(C_s - C) - k_c/U*B0*exp(-k_c*(i-15)/U) - k_n/U*N0*exp(-k_n*(i-15)/U)
end

function simulate_wastewater(C0, B0, N0, distance, dx)
    steps = Int64(distance / dx)
    C = zeros(steps + 1)
    C[1] = C0
    for i in 1:steps
```

```
        if i*dx <= 15
            C[i+1] = C[i] + dx * simulate_timestep1(C[i], i*dx, C0, B0, N0)
        else
            if i*dx == 15
                C0 = C[i]*110/125 + 5*15/125
                B0 = (B0 - B0*exp(-k_c*i/U))*110/125 + 45*15/125
                N0 = (N0 - N0*exp(-k_c*i/U))*110/125 + 45*15/125
            end
            C[i+1] = C[i] + dx * simulate_timestep2(C[i], i*dx, C0, B0, N0)
        end
    end
    return C
end

simulate_wastewater(7.27, 9.09, 7.73, 50, 0.5)

plot(distance_range, simulate_wastewater(7.27, 9.09, 7.73, 50, 0.5), xlabel="Distance downst
plot!(distance_range, DO_values, label="DO Concentration (Analytical)",legend=true)
```



Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Have to adjust the initial concentrations C_0, B_0, N_0 for both control volumes before Stream 2 is added and after. Provided that there is a 5% margin of safety, the absolute limit is 3.8 mg/L. We'll start with the scenario when waste is treated by 50%.

Initial concentration of C_0, B_0, N_0 before Stream 2 is added:

$$C_0 = 7.5\frac{mg}{L} \times \frac{10}{11} + 5\frac{mg}{L} \times 0.5 \times \frac{1}{11} \approx 7.05\frac{mg}{L}$$
$$B_0 = 5\frac{mg}{L} \times \frac{10}{11} + 50\frac{mg}{L} \times 0.5 \times \frac{1}{11} \approx 6.82\frac{mg}{L}$$
$$N_0 = 5\frac{mg}{L} \times \frac{10}{11} + 35\frac{mg}{L} \times 0.5 \times \frac{1}{11} \approx 6.14\frac{mg}{L}$$

$$C(x) = C_s(1 - \alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$
$$\alpha_1 = e^{-\frac{0.55\times15}{6}} \approx 0.252$$
$$\alpha_2 = \frac{0.35}{0.55 - 0.35}(e^{-\frac{0.35\times15}{6}} - e^{-\frac{0.55\times15}{6}}) \approx 0.287$$
$$\alpha_3 = \frac{0.25}{0.55 - 0.25}(e^{-\frac{0.25\times15}{6}} - e^{-\frac{0.55\times15}{6}}) \approx 0.235$$

Concentration at x = 15:

$$C(x = 15) = 10(1 - 0.252) + 7.05(0.252) - 6.82(0.287) - 6.14(0.235) \approx 5.85\frac{mg}{L}$$

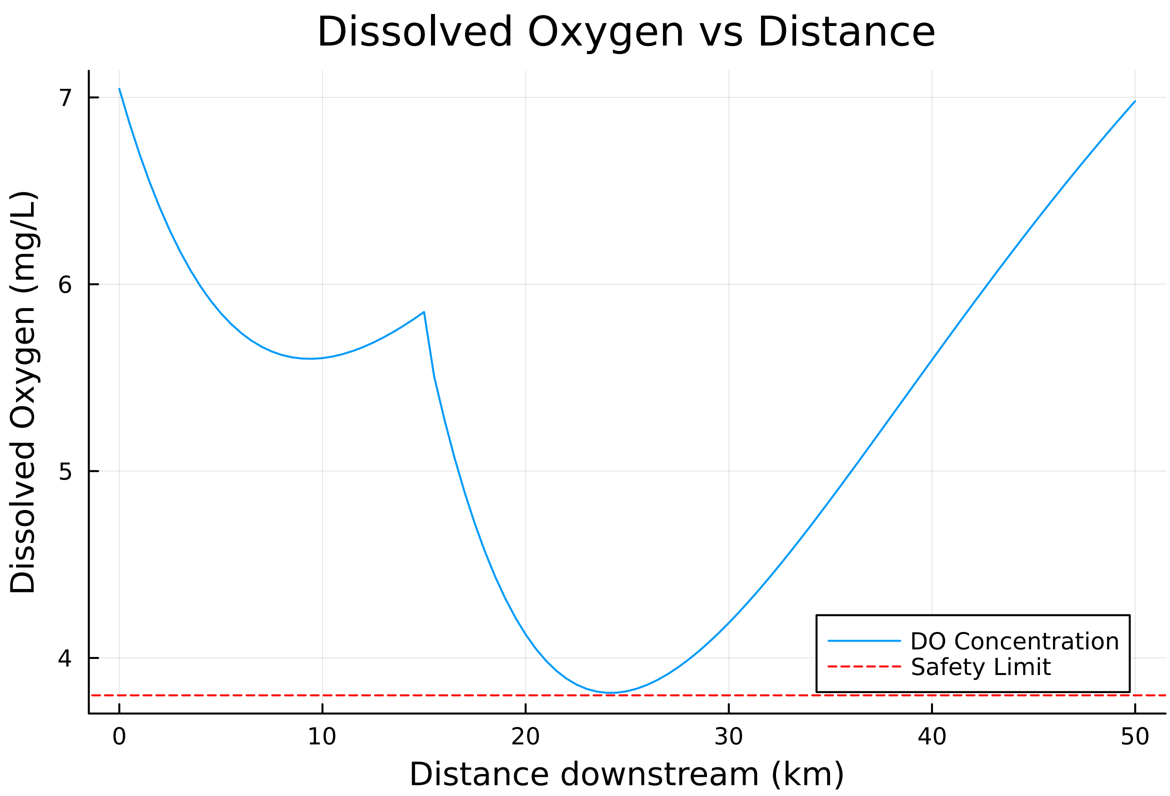
Initial concentration of C_0, B_0, N_0 after Stream 2 is added, assume that Stream 2 is not changed:

$$C_0 = 5.85\frac{mg}{L} \times \frac{110}{125} + 5\frac{mg}{L} \times \frac{15}{125} \approx 5.75\frac{mg}{L}$$
$$B_0 = 6.82(1 - 0.287)\frac{mg}{L} \times \frac{110}{125} + 45\frac{mg}{L} \times \frac{15}{125} \approx 9.68\frac{mg}{L}$$
$$N_0 = 6.14(1 - 0.235)\frac{mg}{L} \times \frac{110}{125} + 35\frac{mg}{L} \times 0.5 \times \frac{15}{125} \approx 8.79\frac{mg}{L}$$

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 0.5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 0.5 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 0.5 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
```

```
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

```
3.8133967134609676
```

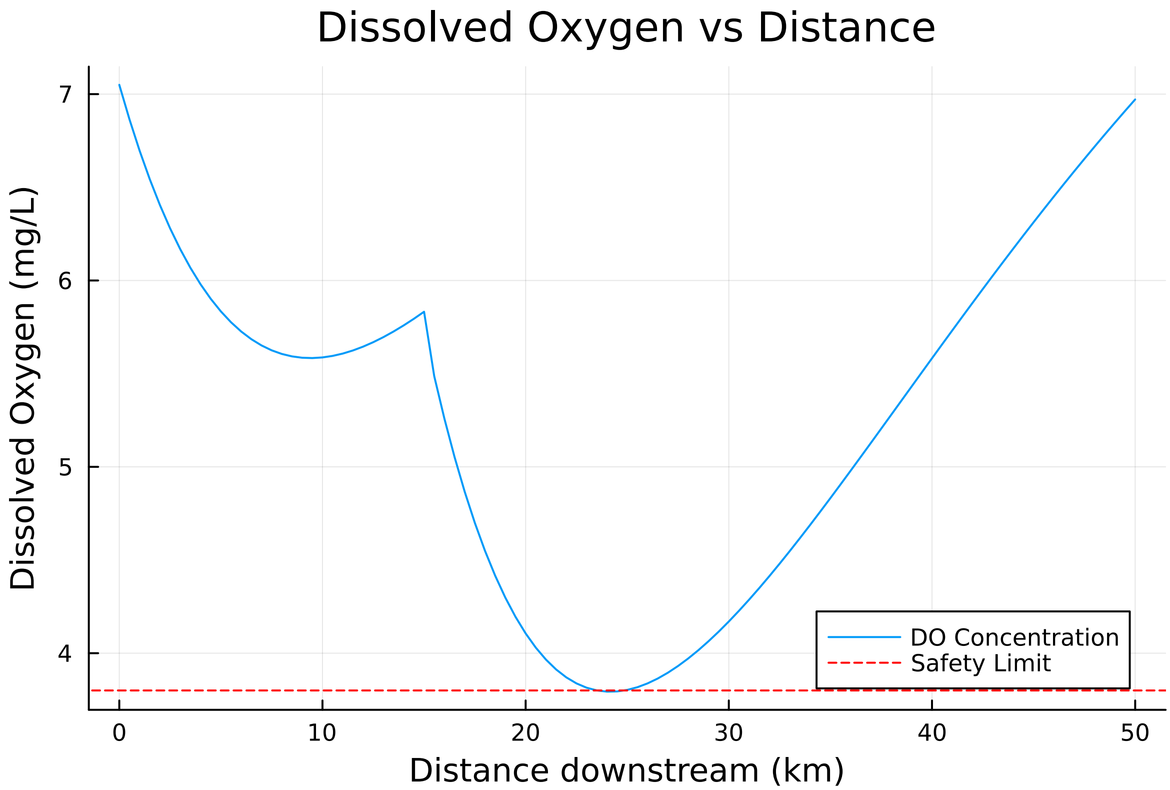


Since this example fortunately skimmed the absolute safety limit in the first try, only one additional test is necessary by confirming if slightly decreasing the treatment % to 49% will make the system still meet the safety limits.

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 0.51 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 0.51 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 0.51 * 1/11 # mg/L
```

```
# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

```
3.7938260915969124
```



As decreasing the % removal of waste to 49% does not meet the absolute safety limit, therefore the minimum percentage of waste to be removed must be at 50%.

For the scenario where only Stream 2 is treated, the initial concentrations C_0, B_0, N_0 and the concentration at x = 15 will be the same as Problem 1.1:

$$C_0 = 7.5\frac{mg}{L} \times \frac{10}{11} + 5\frac{mg}{L} \times \frac{1}{11} \approx 7.27\frac{mg}{L}$$

$$B_0 = 5\frac{mg}{L} \times \frac{10}{11} + 50\frac{mg}{L} \times \frac{1}{11} \approx 9.09\frac{mg}{L}$$

$$N_0 = 5\frac{mg}{L} \times \frac{10}{11} + 35\frac{mg}{L} \times \frac{1}{11} \approx 7.73\frac{mg}{L}$$

Concentration at x = 15:

$$C(x = 15) = 10(1 - 0.252) + 7.05(0.252) - 6.82(0.287) - 6.14(0.235) \approx 5.85\frac{mg}{L}$$

Initial concentrations C_0, B_0, N_0 after Stream 2 is mixed in will be different based on the amount treated. As a starter, a reduction of 50% will be conducted:

$$C_0 = 5.85\frac{mg}{L} \times \frac{110}{125} + 5\frac{mg}{L} \times \frac{15}{125} \times 0.5$$

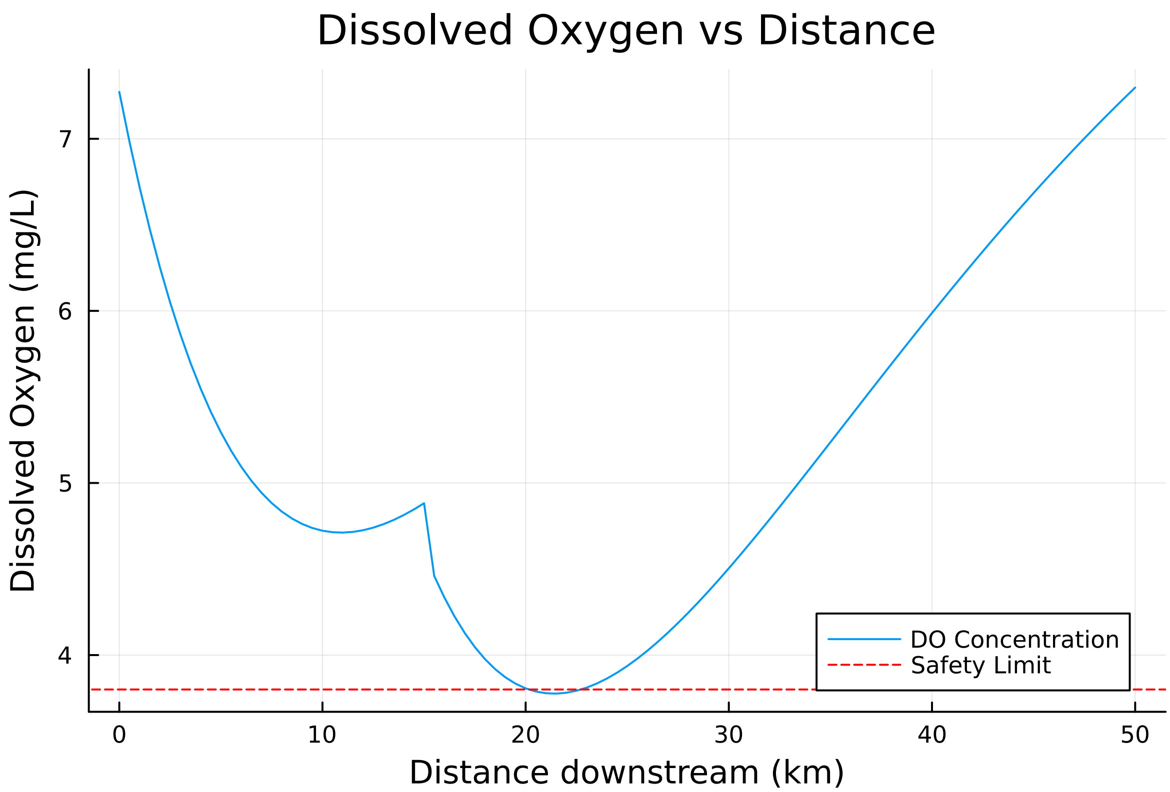
$$B_0 = 9.09(1 - 0.287)\frac{mg}{L} \times \frac{110}{125} + 45\frac{mg}{L} \times \frac{15}{125} \times 0.5$$

$$N_0 = 7.73(1 - 0.235)\frac{mg}{L} \times \frac{110}{125} + 35\frac{mg}{L} \times \frac{15}{125} \times 0.5$$

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.5 # mg
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.5 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 *0.5 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end
```

```
minimum(DO_values)
println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

```
3.7760111801567486
```

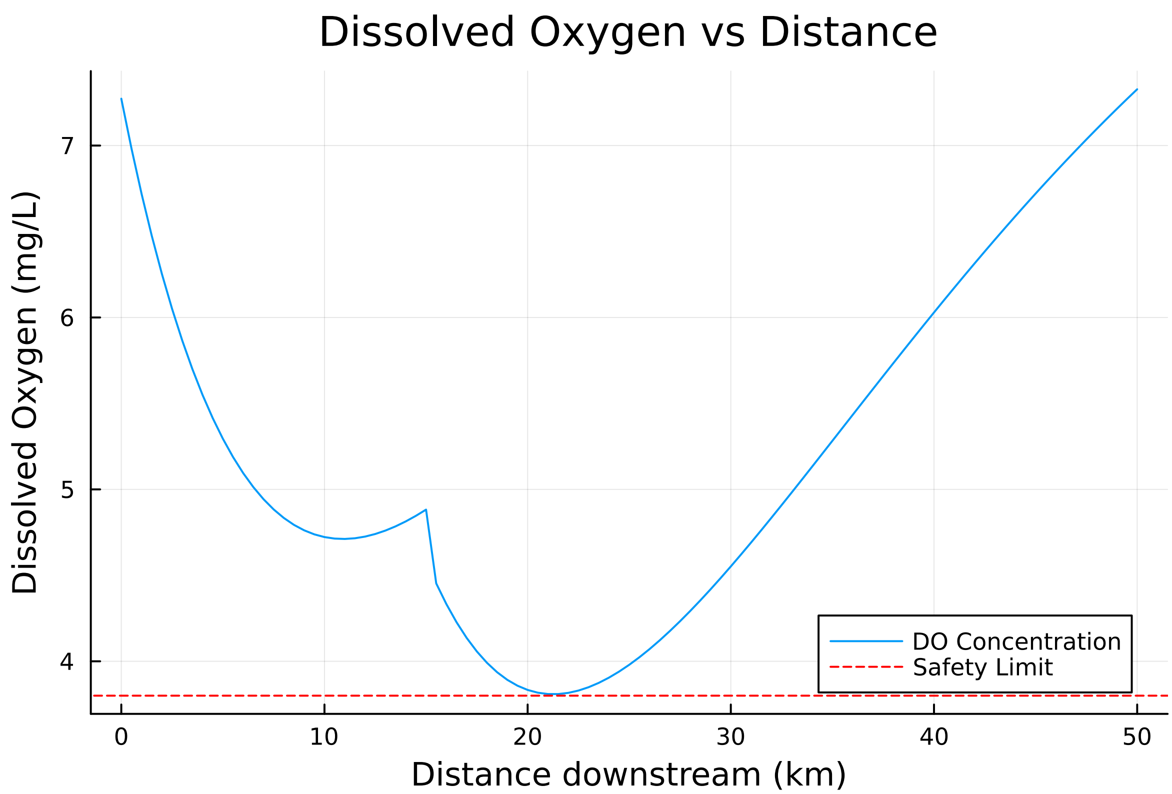


From this simulation, it appears that the system only drops below the safety limit for a small period. Therefore, an additional trial will be conducted where the minimum waste reduced is increased by 52%.

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d
```

```
# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.48 # m
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.48 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 *0.48 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

```
3.809411846219514
```



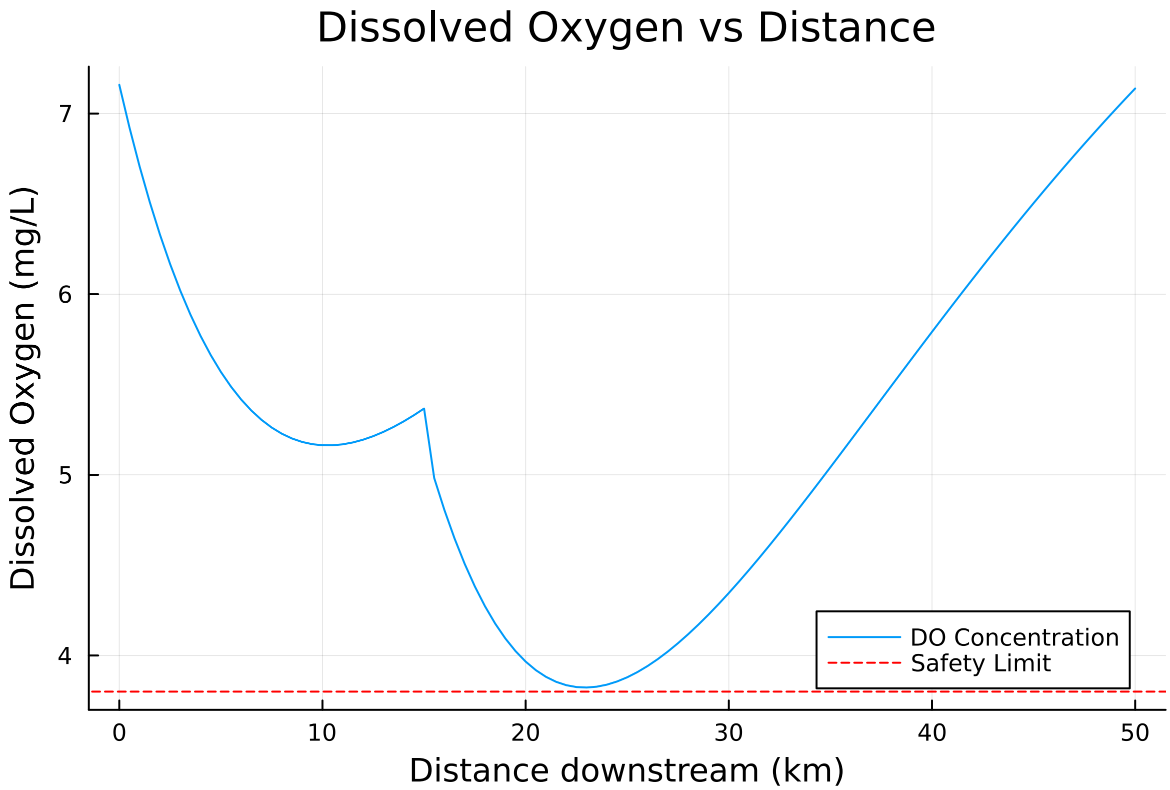
Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 * 0.75 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 * 0.75 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 * 0.75 # mg/L
```

```
# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.75 # m
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.75 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 *0.75 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted /
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted /

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println(minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

```
3.8226341788166613
```



Based on the simulations, it is preferable to treat Stream 2 alone. Since the percentage of waste removed from either stream has similar effects on the magnitude and distance that the minimum value occurs, other considerations about how DO changes through the river should be analyzed. One consideration is the difference in magnitude that DO concentration changes when Stream 2 is added when only Stream 2 is treated vs both streams. When Stream 1 is treated, the DO concentration remains at higher levels from 0km to 15km, but drops significantly after Stream 2 is added. This drastic change in DO could be harmful as it could affect the behaviours of aquatic organisms at that point in the river, even if DO eventually increases later downstream.

Some other information that may be a factor in making a conclusion could be financial and energy costs of treating each stream. Since the effects of removing waste from each stream is similar, if one method of treatment ends up costing less than the other, then that method would be preferred.

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a $\text{LogNormal}(2.0, 0.15)$ distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

```
DO_dist = LogNormal(2.0,0.15)
DO_samples = rand(DO_dist, 1000)
```

```
1000-element Vector{Float64}:
  8.490112929678949
  6.322567297653379
  6.626578909560103
  6.532193283413783
  6.901879890854272
  7.744993368000658
  7.062676758681859
  8.332036226336895
  7.152103170221506
  6.425807315186756
  
  7.721231012133272
  7.524704540111254
 11.831130745924174
  8.48597522938509
  6.9543367202341315
  8.494744132278333
  8.464585443898855
  7.590343331045933
  8.228968823951805
```

References

List any external references consulted, including classmates.